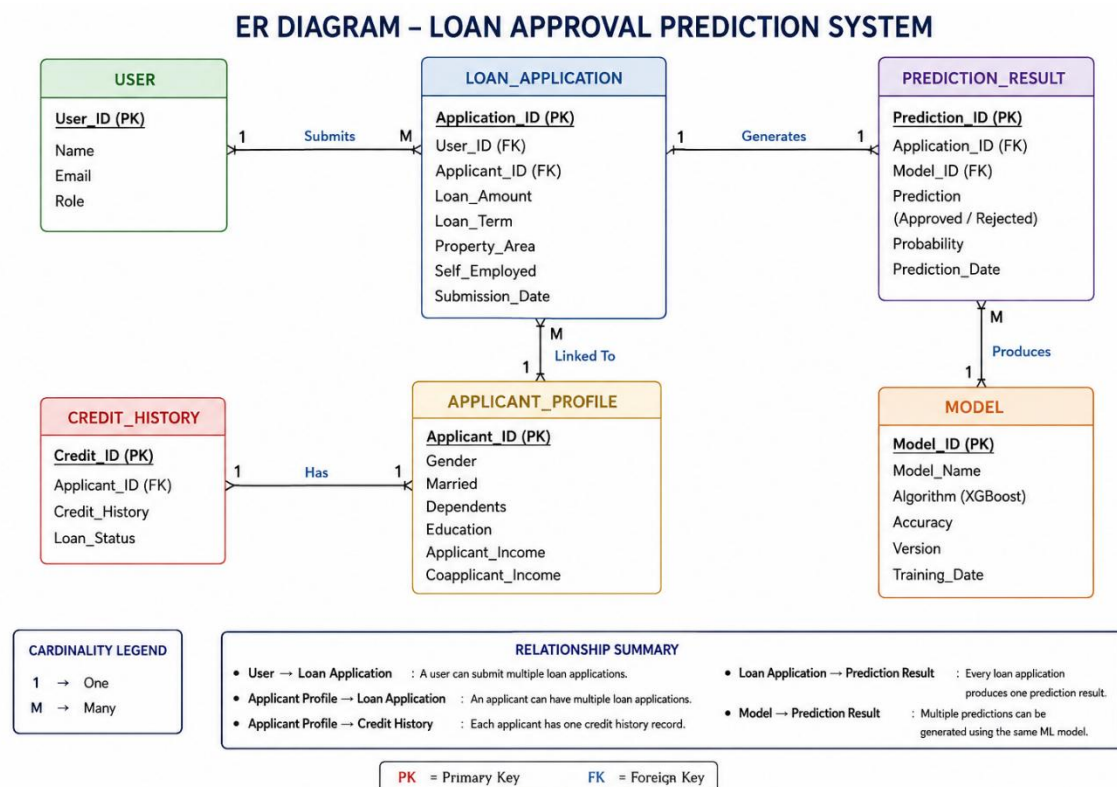


Entity Relationship (ER) Diagram

The Entity Relationship (ER) Diagram illustrates the logical structure of the Smart Lender Loan Approval Prediction System. It identifies the main entities involved in the application, their attributes, and the relationships between them. The ER diagram provides a clear representation of how applicant information, loan applications, credit history, prediction results, and the machine learning model are interconnected. This design helps ensure efficient data organization and supports accurate loan approval prediction using the trained XGBoost model.



ER Diagram Description:

The ER diagram consists of six major entities: **User**, **Applicant Profile**, **Loan Application**, **Credit History**, **Model**, and **Prediction Result**. A user can submit multiple loan applications, while each applicant has a single profile and one corresponding credit history record. Every loan application is processed by the trained machine learning model to generate a prediction result indicating whether the loan is **Approved** or **Rejected**. The same XGBoost model can be used to generate predictions for multiple loan applications. This database structure ensures proper data management, maintains relationships between entities, and supports efficient loan approval prediction within the Smart Lender system.